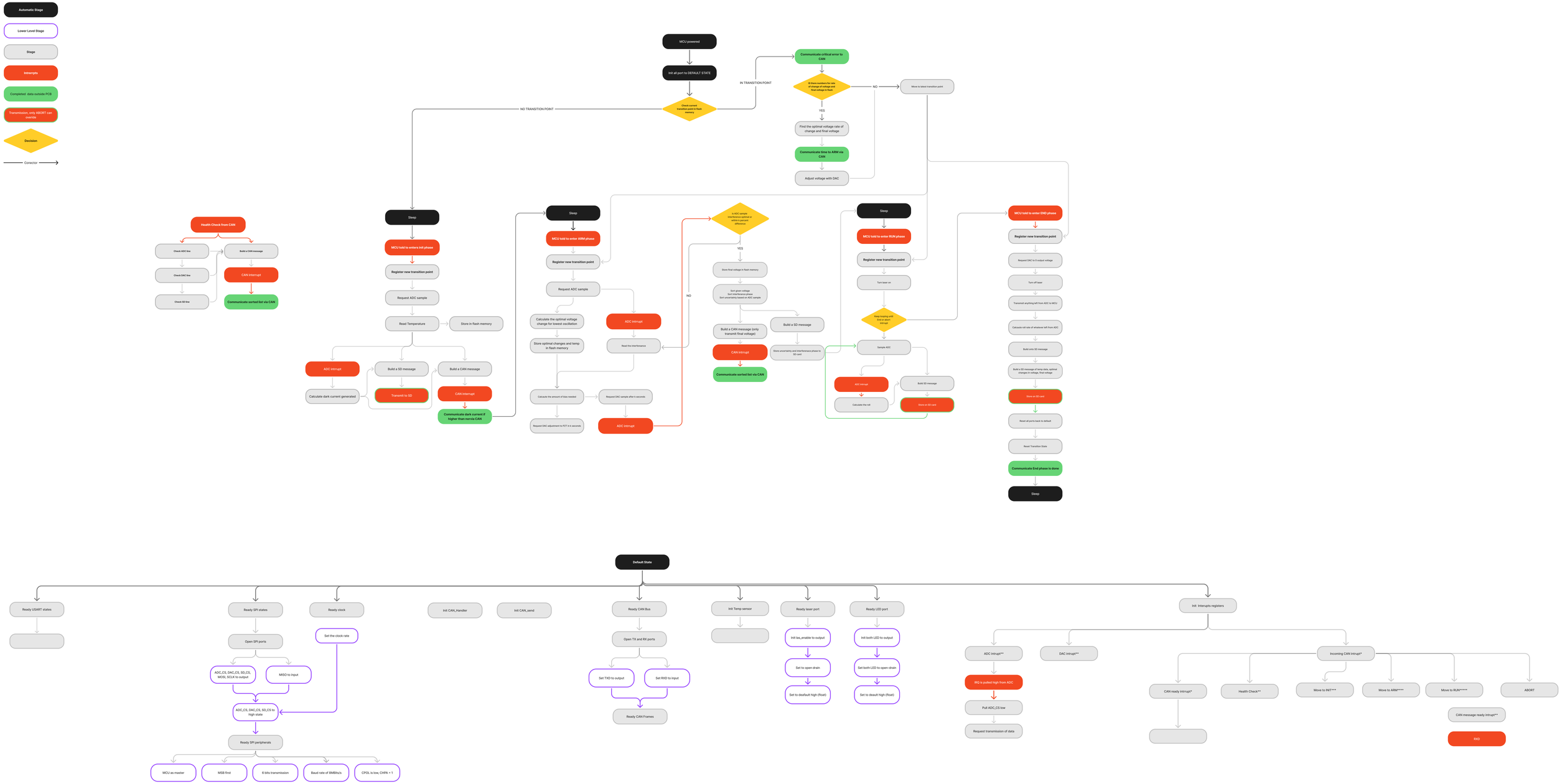
Automatic Stage
Lower Level Stage
Stage
Interrupts
Completed data outside PCB
Transmission, only ABORT can override
Decision
Conector
MCU powered
Init all port to DEFAULT STATE
Check current transition point in flash memory
NO TRANSITION POINT
IN TRANSITION POINT
Communicate critical error to CAN
If there contain for loss of change of voltage and final voltage in flash
NO
YES
Move to latest transition point
Find the optimal voltage rate of change and final voltage
Communicate time to ARM via CAN
Adjust voltage with DAC
Health Check from CAN
Check ADC line
Build a CAN message
Check DAC line
CAN interrupt
Check SD line
Communicate sorted list via CAN
Sleep
MCU told to enters init phase
Register new transition point
Request ADC sample
Read Temperature
Store in flash memory
ADC interrupt
Build a SD message
Build a CAN message
Calculate dark current generated
Transmit to SD
CAN interrupt
Communicate dark current if higher than norvia CAN
Sleep
MCU told to enter ARM phase
Register new transition point
Request ADC sample
Calculate the optimal voltage change for lowest oscillation
ADC interrupt
Store optimal changes and temp in flash memory
Read the interferance
Calcuate the amount of bias needed
Request DAC sample after 5 seconds
Request DAC adjustment to PZT in x seconds
ADC interrupt
Is ROI sample interferance optimal or within 5 percent difference
NO
YES
Store final voltage in flash memory
Sort given voltage
Sort interferance phase
Sort uncertainty based on ADC sample
Build a CAN message (only transmit final voltage)
Build a SD message
CAN interrupt
Store uncertainty and interferance phase to SD card
Communicate sorted list via CAN
Sleep
MCU told to enter RUN phase
Register new transition point
Turn laser on
Keep looping until End or abort Interrupt
Sample ADC
ADC interrupt
Calculate the roll
Build SD message
Store on SD card
MCU told to enter END phase
Register new transition point
Request DAC to 0 output voltage
Turn off laser
Transmit anything left from ADC to MCU
Calcuate roll rate of whatever left from ADC
Build onto SD message
Build a SD message of temp data, optimal changes in voltage, final voltage
Store on SD card
Reset all ports back to default
Reset Transition State
Communicate End phase is done
Sleep
Default State
Ready USART states
Ready SPI states
Open SPI ports
ADC_CS, DAC_CS, SD_CS, MOSI, SCLK to output
MISO to input
ADC_CS, DAC_CS, SD_CS to high state
Ready SPI peripherals
MCU as master
MSB first
6 bits transmission
Baud rate of 8MBits/s
CPOL is low, CHPA = 1
Ready clock
Set the clock rate
Init CAN_Handler
Init CAN_send
Ready CAN Bus
Open TX and RX ports
Set TXD to output
Set RXD to input
Ready CAN Frames
Init Temp sensor
Ready laser port
Init las_enable to output
Set to open drain
Set to deafault high (float)
Ready LED port
Init both LED to output
Set both LED to open-drain
Set to deault high (float)
Init Interrupts registers
ADC intrupt**
IRQ is pulled high from ADC
Pull ADC_CS low
Request transmission of data
DAC intrupt**
Incoming CAN intrupt*
CAN ready intrupt*
Health Check**
Move to INIT***
Move to ARM****
Move to RUN*****
ABORT
CAN message ready intrupt**
RXD